

MIHIR MITHANI

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PROFESSIONAL SUMMARY

B.Tech ICT student (CGPA: 8.88) at Marwadi University with a strong foundation in algorithms, data structures, and systems software, GPU-accelerated and high-performance computing, and deep learning frameworks (CUDA, TensorRT, Triton Inference Server, NVIDIA DALI, TAO Toolkit). Skilled in C/C++ and Python with hands-on experience analyzing and optimizing large-scale workloads, deploying models for scaled inference, and researching data-driven performance improvements. Selected for the IEEE Campus Immersion Program at IIT Guwahati covering supercomputing facilities and quantum computing. Seeking a Software Performance at Scale internship to help develop, evaluate, and optimize software performance across large GPU clusters.

SKILLS

- Languages & Algorithms: C, C++, Python, MATLAB, Verilog HDL; strong foundations in Data Structures & Algorithms (DSA)
- GPU-Accelerated & Deep Learning Frameworks: CUDA, NVIDIA TensorRT, Triton Inference Server, NVIDIA DALI, TAO Toolkit, NVIDIA CuOpt & NIM, LightGBM
- High-Performance & Large-Scale Computing: Multi-Core Computer Architecture, GPU cluster workload analysis, scaled inference deployment, performance benchmarking
- Robotics & Simulation: ROS, NVIDIA IsaacLab, IsaacSim, USD Composer, Reinforcement Learning
- Hardware & Embedded Platforms: Arduino Uno R4 WiFi, NVIDIA Jetson Nano, Raspberry Pi, EasyEDA (PCB Design), Wokwi
- Dev Tools & Cloud: Git/GitHub, Flask, Gradio, Firebase, Blynk Cloud

EXPERIENCE

VLSI Intern | **Microcircuits Innovations Pvt. Ltd.** | Jun 2026 – Present

- Training and working on VLSI Backend design, gaining hands-on experience with back-end VLSI flow including physical design, timing analysis, and layout implementation.

Project Coordinator & Team Leader | **Robotics Team (ICT), Marwadi University** | Sep 2025 – Present

- Led a 15-member team to design and deliver industry-grade robotics projects (SO100 Arm and TurtleBot) using NVIDIA frameworks including ROS, IsaacLab, IsaacSim, USD Composer, and Reinforcement Learning.
- Trained RL policies in IsaacLab, evaluated performance in IsaacSim, and deployed to physical hardware, bridging the sim-to-real gap; built and benchmarked ROS-based communication pipelines, cutting operational costs by 30-40% at scale.

Hardware Intern (Full-time) | **Einfochips** | Jun 2025 – Aug 2025

Remote

- Designed and validated an IoT-based hardware prototype improving restaurant operations through integrated sensor modules and relay controls.
- Analyzed and streamlined operational workflows and debugging pipelines, identifying inefficiencies and improving system performance by 30%+.

PROJECTS

Disaster Risk Monitoring System Using Satellite Imagery *Deep Learning · Performance Optimization · NVIDIA* [\[GitHub\]](#)

- Developed a Sentinel-1 SAR-based deep learning framework for flood detection using U-Net semantic segmentation (0.85 mIoU); accelerated large-scale data preprocessing with NVIDIA DALI and trained with TAO Toolkit.
- Optimized inference performance by deploying TensorRT-optimized models on Triton Inference Server (~40 images/sec at scale); geospatial impact analysis identified ~195,000 potentially affected individuals in real-world flood scenarios.

GarbageBot – RL Control Center (OpenEnv) *Reinforcement Learning · LLM · Robotics* [\[GitHub\]](#) [\[Live Demo\]](#)

- Built an OpenEnv-compliant reinforcement learning environment for a garbage-collecting robot, modeling grid navigation, battery management, and storage logistics across tasks ranging from 5x5 to 10x10 grids with increasing obstacle complexity.
- Designed a policy priority chain combining a pre-trained Q-Learning table, a 4-bit quantized fine-tuned Llama-3.2-3B-Instruct policy, and a BFS heuristic fallback; deployed the environment and live dashboard via FastAPI and Docker for real-time policy evaluation.

Chip Routing Optimizer using NVIDIA CuOpt & NIM *VLSI · GPU Optimization · Algorithms · EDA* [\[GitHub\]](#)

- Built a GPU-accelerated chip routing engine applying NVIDIA CuOpt and NIM APIs to formulate and solve VLSI back-end wire-routing as a constrained graph optimization problem.
- Analyzed solver performance and minimized wire length and routing congestion across 3 iterative versions (v1-v3), with progressive algorithmic refinements in net ordering, layer assignment, and congestion avoidance, outperforming classical heuristics.

FaultSense – Industry Equipment Fault Detection *ML · Performance Benchmarking · Flask · Gradio* [\[GitHub\]](#) [\[Live Demo\]](#)

- Built a real-time binary fault detection system for 5 industrial equipment types (Pump, Compressor, Motor, Valve, Sensor) using LightGBM, achieving AUC 0.97, F1 0.93, Recall 0.95 on 10,000 samples.
- Ran a large-scale grid search across ~13,650 model configurations to identify optimal performance trade-offs; deployed as a Flask REST API with a live Gradio web UI on Hugging Face Spaces.

Smart Safety and Control LPG System – Hardware & PCB *IoT · Embedded Systems · PCB Design* [\[GitHub\]](#)

- Designed a complete hardware safety system integrating an MQ-6 LPG gas sensor, R307S fingerprint sensor, relay-controlled solenoid valve, and LED alert on an Arduino Uno R4 WiFi.
- Performed full PCB schematic design in EasyEDA, resolving footprint mismatches and net mapping errors; integrated real-time cloud monitoring via Firebase and validated the design with Wokwi simulation.

Automatic Terrace Farming Irrigation & Growth Monitoring System *IoT · Embedded · Blynk*

- Built a Wi-Fi-enabled robotic irrigation system with real-time plant growth monitoring via Blynk Cloud.

Facial Emotion Recognition *AI/ML · MATLAB · Computer Vision* [\[GitHub\]](#)

- Developed a real-time facial emotion recognition model in MATLAB using the FER2013 dataset with a trained ML classification pipeline.

EDUCATION

Marwadi University – Rajkot, India *2023 – 2027*

B.Tech, Information and Communication Technology | CGPA: 8.88 | NAAC A+

Relevant Coursework: Algorithms & Data Structures, VLSI Design, Digital Design, Embedded Systems, IoT, AI/ML, High-Performance Computing, Robotics

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- **NVIDIA Deep Learning Institute:** Fundamentals of Accelerated Computing with CUDA | AI on Jetson Nano | Disaster Risk Monitoring via Satellite Imagery
- **NPTEL:** Digital Design with Verilog | MultiCore Computer Architecture
- **Intel Learning:** Verilog HDL Basics
- **Google:** Introduction to Large Language Models | Introduction to Generative AI | Introduction to Responsible AI

HONORS & AWARDS

- **IEEE CIP Program – IIT Guwahati:** Selected for a 15-day campus immersion covering supercomputing facilities and quantum computing workshops.
- **CodeJam GFG Winner – GeeksforGeeks:** Recognized for solving complex DSA problems at the GFG hackathon in collaboration with Marwadi University.
- **Best Team Synergy Award 2025 – ICT, Marwadi University:** Awarded for leading the Robotics Team (SO100 Arm & TurtleBot) over a 6+ month engagement.
- **C3 Winner 2024 – MUFEST, Marwadi University:** First rank for an IoT-based smoke detection and alert system with real-time data visualization.
- **ByteHunt IoT Trial 2025 Winner – Marwadi University:** First rank for a gyroscope sensor monitoring solution with cloud-based data visualization.
- **NetSip – Marwadi University:** Attended seminars on Network and Signal Processing organized by IEEE India Council and IEEE-SPS.